

Graphs of Modulus and Reciprocals

Key Points

For a function $f(x)$ to draw the graph

- $|f(x)|$ we reflect the graph in the x -axis for the domain when $f(x) < 0$
- $\frac{1}{f(x)}$ we have asymptotes at x such that $f(x) = 0$ and as we approach these points the function tends to ∞ . Maximums and Minimums switch. If the function tends to infinity then now tends to 0.

Basic Skills

Q1. Sketch the graph of

a) $y = 4x - 2$

b) $y = |4x - 2|$

c) $y = \frac{1}{4x-2}$

Q2. Sketch the graph of

a) $y = x^2 - 8x - 20$

b) $y = |x^2 - 8x - 20|$

c) $y = \frac{1}{x^2-8x-20}$

Q3. Sketch the graph of $y = |\sin x|$

Q4. Sketch the graph of $y = \frac{1}{\cos x}$ (i.e. $y = \sec x$)

Competency Skills

Q5. a) On the same graph sketch $y = |2 - 3x|$ and $y = x + 5$ showing the points of intersection

b) Hence solve the inequality $|2 - 3x| < x + 5$

Q6. a) Sketch the graph $y = |\tan x|$ for $-\pi < x \leq \pi$

b) Hence, or otherwise, solve the inequality $|\tan x| < 0.3$ in the domain $-\pi < x \leq \pi$

Q7. Sketch the graph $y = \frac{1}{\ln x}$

Q8. Sketch the graph of $y = 3 - 2|1 - x|$

Q9. By sketching appropriate functions solve the inequality $|5 - x^2| \geq x$

Q10. Sketch the graph $y = \frac{1}{e^{-x}-5}$

Advanced Skills

Q11. a) On the same graph sketch $y = \frac{x+2}{x-1}$ and $y = \frac{x-1}{x+2}$ showing the points of intersection

b) Hence state when $\frac{x+2}{x-1} < \frac{x-1}{x+2}$

Q12. Sketch the graph $y = \left| \frac{1}{x-5} \right|$

Q13. The graph $y = a|e^{2x} - 3| + b$ passes through the coordinates (5,-5) and (0,2) find the constants a and b and sketch the graph

Q14. Sketch the family of graphs $y = a|\ln(|x|)|$ for various a distinguishing between $a > 0$ and $a < 0$

Q15. Sketch the graph of $y = \frac{1}{|\tan x|}$ using this to determine if $\frac{1}{|\tan x|} \equiv |\cot x|$

Extension

Q16. Draw the rational graph $y = \frac{x^2-5x+4}{2x(x-2)}$

Hint: Plot asymptotes (undefined and limits), both types of intercepts and max/min points

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